Decodable but Not Load-Bearing: Why Physical Inductive Biases Fail in Latent World Models

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Abstract

Latent world models can *decode* physical state from their representations, yet fail to *obey* physical law when predictions are rolled out: a joint-embedding predictive (JEPA-style) model that recovers object position almost perfectly at one step collapses over long horizons and under out-of-distribution (OOD) physics. A popular remedy injects physical inductive biases into the latent space. Testing this remedy on a JEPA world model across five injection mechanisms (fixed physical slots, kinematic dynamics, deep-supervision probes, consistency losses, and label-free priors), two training regimes (post-hoc fine-tuning and from-scratch pretraining), three controlled physics domains, and two photorealistic-simulation benchmarks, we find it consistently *hurts*, with damage growing as the baseline strengthens (up to $+0.56$ normalized MSE from scratch). We trace the failure to a *load-bearing problem*: physical dimensions occupy a vanishing fraction of the latent and its gradients, so prediction routes around them; a one-line loss reweighting removes the harm where the slot matches the dynamics, but never yields a net gain. The variables that actually govern physical robustness lie in training and data, not structure: autoregressive free rollout is the single dominant lever (2.2–$3.6\times$ OOD error reduction across all domains, seed-robust, and the best transfer to photorealistic scenes), rollout horizons must match dynamics complexity (a $19\times$ long-horizon gain on photorealistic video when paired with geometric scale jitter), and augmentation must match the domain's true nuisances—appearance jitter, the strongest lever on simple synthetic data, *reverses* catastrophically (up to $100\times$) on photorealistic scenes where appearance carries physical information. Finally, we show that cosine- and probe-based metrics are duals of the auxiliary training losses and systematically overstate physical competence, explaining why structural remedies can appear to work.

1 Introduction

World models promise agents an internal simulator: given a few frames, they should predict how objects move, fall, and collide (Ha and Schmidhuber 2018; LeCun 2022). Current latent world models fall short of this promise in a specific way—not by failing to *represent* physical state, but by failing to *evolve* it. On the PhyWorld benchmark (Kang et al. 2024), a JEPA-style world model (Maes 2025; Balestriero and LeCun 2025) encodes object position and velocity almost perfectly (linear-probe correlation up to 0.96), and its one-step predictions are near-exact (latent cosine 0.98–0.99). Yet when the same model is rolled out autoregressively, prediction error compounds: over 28 steps the latent cosine on collision dynamics falls to 0.24, and out-of-distribution (OOD) physics—unseen radii, masses, or velocities—multiplies rollout error several-fold over the in-distribution level. State decodability does not imply physical-law compliance.

Two families of remedies dominate the literature. The first injects *physical structure* into the representation: binding latent dimensions to physical quantities, attaching kinematic update equations, or supervising physics through auxiliary probes—an approach exemplified by physically interpretable world models (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025) and deep-supervision methods (Zahorodnii 2025), which report improved decodability and reduced drift. The second scales *data and training*, a route whose limits Kang et al. (2024) documented for video generation: diffusion-based generators trained at scale still fail OOD physics, generalizing case-by-case rather than by rule. Whether physical structure actually rescues *latent* world models—where prediction happens in representation space rather than pixel space—has not been tested systematically.

We study this question in a controlled setting. A latent world model consists of an encoder mapping frames to embeddings and a predictor rolling embeddings forward, trained by matching predicted to encoded futures (JEPA). *Teacher forcing* trains the predictor on one-step transitions from ground-truth context; *free rollout* trains it on its own multi-step predictions (Bengio et al. 2015; Venkatraman, Hebert, and Bagnell 2015). PhyWorld provides three single-mechanism physics domains—uniform motion, parabolic flight, and elastic collision—with explicit in-distribution ranges over radius/mass and velocity, yielding four evaluation partitions (ID, r/m-OOD, v-OOD, both-OOD). We report normalized MSE (nMSE) in latent space and PSNR of decoded rollouts in pixel space, and we deliberately avoid cosine similarity and probe correlation as primary metrics, for reasons the paper establishes (§6).

Existing evidence for physical structure rests on fragile ground. The gains these methods report are typically read from probe correlation or latent cosine—quantities we show to be *duals of the auxiliary training loss itself*, which rise mechanically as that loss is optimized regardless of whether prediction improves. Injection mechanisms are evaluated in isolation, on single domains, under a single initialization regime, and rarely against the confound that dominates everything else: the training protocol. No study answers the basic factorial question—does physical structure help once training is done right?

We answer it by exhaustive, controlled anatomy. On a fixed backbone we cross five injection mechanisms (fixed slots, kinematic heads with both free and exact gravitational forms, deep-supervision probes, velocity-consistency losses, and label-free dynamics priors) under two training regimes (post-hoc fine-tuning and from-scratch co-training), three physics domains, and seed-controlled baselines; we then validate the surviving conclusions on photorealistic simulation (Physion and Physion++ (Bear et al. 2021; Tung et al. 2023), rendered in the ThreeDWorld engine). We further propose a mechanistic explanation—the *load-bearing problem*: physical slots occupy 2 of 192 latent dimensions and $\sim 1\%$ of the prediction loss, the black-box channel redundantly encodes position, and auxiliary gradients conflict with prediction gradients (probe-to-prediction gradient ratios of 15–$125\times$ collapse the encoder's effective dimensionality). The explanation makes a falsifiable prediction: reweighting the loss to make slots load-bearing should remove the damage. On uniform motion it does—restoring parity with the baseline (0.132 ± 0.017 vs. 0.136 ± 0.009 both-OOD nMSE)—but never more, and on richer dynamics even the reweighted slot stays harmful.

Our results separate cleanly. Physical structure fails wherever it can be trusted: across 30 mechanism–domain cells, none delivers a seed-robust improvement and most degrade prediction outright, and the damage *grows* as baselines strengthen—with a clean from-scratch baseline, adding physics costs $+0.56$ nMSE on uniform motion, an order of magnitude more than the $+0.035$ cost of post-hoc grafting, closing the "structure must be baked in during pretraining" escape hatch. Training protocol succeeds everywhere: replacing teacher forcing with free rollout cuts OOD error 2.2–$3.6\times$ across all three domains (seed-robust, non-overlapping intervals); matching the rollout horizon to dynamics complexity yields further gains, up to $19\times$ on long-horizon photorealistic video when paired with geometric scale jitter; and domain-matched augmentation halves OOD error on simple synthetic data—yet appearance jitter, the strongest such lever, *reverses* on photorealistic video, degrading friction-collision prediction $100\times$.

This paper contributes:

- the first systematic, seed-controlled anatomy of physical inductive biases in latent world models, showing consistent failure across five mechanisms, two training regimes, three synthetic domains, and two photorealistic-simulation benchmarks (§4);
- a mechanistic account—the load-bearing problem—validated by a falsifiable reweighting intervention (§4.3);
- a training-and-data recipe (free rollout, horizon–complexity matching, domain-matched augmentation) with an explicit simple-synthetic-to-photorealistic boundary (§5);
- an evaluation methodology exposing four systematic metric failure modes, including cosine/probe duality with the training loss and nMSE denominator blow-ups, with per-partition, per-horizon cross-validation rules (§6).

2 Related Work

Physical reasoning benchmarks for world models. PhyWorld (Kang et al. 2024) generates controlled single-mechanism physics videos (uniform motion, parabola, collision) with explicit in-distribution parameter ranges, and showed that scaled video-diffusion generators memorize cases rather than laws, failing under OOD physics. Physion (Bear et al. 2021) and Physion++ (Tung et al. 2023) test object-contact prediction and property inference on realistic simulated video. We connect these two worlds: we use PhyWorld's OOD protocol to dissect *latent* (JEPA-style) world models rather than pixel-space generators, and we validate every surviving conclusion on Physion/Physion++. Our finding that zero-shot transfer from synthetic training is capped by the random-encoder prior extends the benchmark literature with a cautionary calibration: architecture priors alone set a floor that synthetic pretraining fails to exceed.

Physically structured world models. A growing line of work argues that world models should carry explicit physical state: physically interpretable world models align latent dimensions with physical quantities through weak supervision (Mao, Umasudhan, and Ruchkin 2024), design principles for such models are catalogued by Peper et al. (2025), and deep supervision with linear probes injects physical readouts into the training loss (Zahorodnii 2025). Hamiltonian neural networks impose conservation structure directly on the dynamics (Greydanus, Dzamba, and Yosinski 2019). These works report improved interpretability and reduced drift, typically measured by probe accuracy or latent similarity. We test the transferable core of these proposals—fixed slots, kinematic equations (including the exact gravitational form), probe supervision, and their label-free variants—inside a shared-latent JEPA model, and find all of them harmful under scale-aware metrics. Our analysis attributes the discrepancy to architecture: extrinsic designs make the physical state load-bearing by construction, whereas grafting the same ingredients onto a shared latent leaves them decorative (§4.3).

Rollout training and exposure bias. Training autoregressive predictors on their own outputs is a classic remedy for compounding error: scheduled sampling (Bengio et al. 2015), Data-as-Demonstrator (Venkatraman, Hebert, and Bagnell 2015), and professor forcing (Goyal et al. 2016) all attack the train–test mismatch of teacher forcing, and modern latent world models train on imagined rollouts (Ha and Schmidhuber 2018; Hafner et al. 2023). We claim no novelty for free rollout itself. Our contribution is comparative: under identical, seed-controlled conditions, this training-protocol choice dominates *every* physical-structure intervention across both synthetic and photorealistic-simulation data, and we quantify a horizon–complexity matching law that assigns each domain its optimal rollout length.

Data augmentation for dynamics OOD. Physics-informed augmentation has been proposed for vehicle dynamics (Maheshwari et al. 2023) and world-model robustness (Ball et al. 2021). On synthetic domains we confirm augmentation as the strongest OOD lever—appearance jitter halves the hardest-split error—but we document two boundaries this literature has not: augmentation interactions are non-monotone (appearance conflicts with long-horizon training while geometric scaling synergizes with it), and appearance-based gains reverse catastrophically on photorealistic simulation, where appearance carries physical information rather than acting as nuisance. Together with the JEPA representation-learning line (Assran et al. 2023; Bardes et al. 2024; Balestriero and LeCun 2025), these results position our study as a systematic empirical audit of what actually confers physical robustness in latent world models.

3 Experimental Setup

Model. We use LeWM (Maes 2025), a compact JEPA-style action-conditioned world model ($\sim$10M parameters): a ViT-tiny encoder E_θ producing a 192-dimensional embedding, an MLP projector, a lightweight action encoder, and a 6-layer causal-transformer predictor F_ϕ. Training minimizes the latent prediction error $\|F_\phi(\mathbf{z}_{1:t}) - E_\theta(o_{t+1})\|^2$ plus a SIGReg anti-collapse regularizer (Balestriero and LeCun 2025). Unless stated otherwise, the encoder is initialized from the public PushT-pretrained checkpoint; §4 additionally studies from-scratch training. This small, fully controlled backbone lets us run the full factorial anatomy (over 60 training runs) and repeat each headline configuration across three random seeds, so that reported effects can be separated from seed-to-seed noise; we discuss external validity in §7.

Training regimes. The predictor can be trained two ways. *Teacher forcing* (LeWM's default) trains one-step transitions from ground-truth context ($\mathrm{num\_preds}=1$). *Free rollout* feeds the predictor its own predictions autoregressively for H steps ($\mathrm{num\_preds}=H$, default 8) and supervises the whole rollout, exposing the model to its own compounding errors during training. Because free rollout is the dominant lever we identify (§5.1), it is our default throughout; all physical-structure variants are added on top of it, and teacher forcing appears only as the explicit baseline it improves upon.

Domains and OOD protocol. PhyWorld (Kang et al. 2024) provides three single-mechanism domains: *uniform motion* (zero acceleration), *parabola* (constant gravity), and *collision* (elastic impulse; discontinuous acceleration). Each model trains on the $1{,}000$ in-distribution trajectories that define one mechanism (32 frames, 224×224; ID means radius/mass $\in [0.7, 1.5]$ and velocity $\in [1, 4]$) and is evaluated on four partitions: ID, r/m-OOD (radius/mass out of range; changes appearance), v-OOD (velocity out of range), and both-OOD (both out of range, the hardest split). The small, clean training set is deliberate: it follows PhyWorld's controlled protocol and isolates the inductive-bias question from data-scale confounds, so that any effect we measure is attributable to the injected structure rather than to data volume. We separately confirm the surviving conclusions at realistic scale below.

Photorealistic-simulation benchmarks. Physion (Bear et al. 2021) and Physion++ (Tung et al. 2023) are rendered in ThreeDWorld, a Unity-based engine that produces photorealistic—not camera-captured—video with rich multi-object dynamics far beyond PhyWorld's single-mechanism scenes; they are our realism stress test. We use them in the two standard ways. On Physion we evaluate *zero-shot transfer*: a PhyWorld-trained encoder is frozen and a linear object-contact-prediction (OCP) readout is fit over eight scenarios, testing whether synthetic-domain training yields features that carry to photorealistic scenes without further training. On Physion++ we instead *train directly* on its readout clips (which supply ground-truth object states) and evaluate latent rollout up to horizon 64, testing long-horizon prediction within the realistic-simulation regime itself.

Metrics and protocol rules. Our primary metrics are scale-aware: *latent nMSE*, $\|\hat{\mathbf{z}} - \mathbf{z}\|^2$ normalized by the variance of the target latents (0 is perfect; $\approx$1 matches predicting the mean), and *pixel PSNR* of rollouts decoded by a universal decoder trained per model (position dominates the visual error, making this the sharpest lens for physical accuracy; collision is excluded from pixel evaluation because its decoder is reconstruction-limited). As diagnostics only, we report latent cosine and probe decodability ρ (ridge readout of position/velocity from rolled-out latents; velocity uses $K{=}4$ stacked frames). All numbers are reported per partition *and* per horizon; §6 shows why aggregate or direction-only metrics mislead, and why the parabola both-OOD aggregate specifically must be replaced by its clean r/m-OOD partition. Data splits are trajectory-level. Headline comparisons use three seeds (mean±std); remaining cells are single-seed and marked accordingly.

4 Physical Inductive Biases Fail

We inject physical structure into the shared 192-dimensional latent through five mechanisms spanning the design space—from pinning *what the state is* to constraining *how it evolves*, with and without labels (Table 1). All variants run on top of free rollout, i.e., they must beat an already strong training protocol, which is the deployment-relevant question.

4.1 Post-hoc Injection: Every Mechanism Hurts

Table 2 reports the hardest clean-split latent nMSE for each mechanism against the pure free-rollout baseline—30 mechanism–domain cells in total. Three observations stand out. First, *correct physics does not help*: on parabola, the strict gravitational head—the ground-truth functional form, with only the gravity constant learned—degrades error from 0.127 to 0.173, no better than an unconstrained MLP head (0.178). Second, *labels do not help*: the label-free prior degrades all three domains (collision most, $0.393 \to 0.653$), and so does its fully grounded counterpart with perfect position labels ($0.131 \to 0.166$ on uniform; $0.393 \to 0.525$ on

Mechanism	Constrains	Labels	Evol. form
Fixed slot	what state *is*	yes	—
Probe	what state *is* (read-out)	yes	—
Kinematic head	how state *evolves*	yes	a=0, g / MLP
Consistency	how state *evolves*	yes	none
Label-free prior	how state *evolves*	**no**	2nd-order

Table 1: Five physical-structure injection mechanisms, spanning constraints on *what* the state is to constraints on *how* it evolves, with and without labels. The "Evol. form" column applies only to evolution mechanisms: the kinematic head uses a fixed acceleration form—a=0 (constant velocity) or the exact gravitational a=g with one learnable parameter (strict PIWM)—or a free MLP; consistency imposes no acceleration form; the label-free prior only requires smooth second-order evolution, with no ground-truth grounding.

collision). Third, *combinations do not rescue*: probe supervision alone is harmful ($0.131 \rightarrow 0.167$ on uniform, confirmed in pixel space, $20.41 \rightarrow 19.83$ dB; $0.393 \rightarrow 0.647$ on collision), and adding it to the reweighted slot yields parity at best (0.125 vs. 0.132 ± 0.017 on uniform; 0.127 on parabola) or clear damage (0.607 on collision), while its pixel rollout falls below even the baseline (20.02 vs. 20.41 dB). The form-free consistency loss, designed specifically for collision impulses, fails worst exactly there (0.57–0.64 vs. 0.393), and fails on the smooth domains too (0.151 vs. 0.131; 0.147 vs. 0.127).

Across the grid, 28 of 30 cells fail to improve on the baseline, and the two apparent exceptions—both single-seed, both on parabola's r/m-OOD split—do not survive scrutiny. Probe-only parabola reads 0.115, exactly the bottom of the baseline's own seed range (0.115–0.127): seed noise. The slot carrying position+velocity reads 0.093 on parabola while being the *worst* uniform variant ($0.114 \rightarrow 0.207$) and clearly harmful on collision (0.621); we replicate this single cell across seeds before reading anything into it, and note that a mechanism whose sign flips across the two smooth domains cannot support a robustness claim in any case.

4.2 From-Scratch Co-Training Hurts More

A natural defense holds that physical structure cannot be grafted onto an encoder already committed to a black-box representation—it must be baked in during pretraining, as extrinsic architectures do. We test this with a 2×2 factorial (init $\in$ {PushT post-hoc, from-scratch} $\times$ physics $\in$ {off, on}) per domain, with learning rate re-tuned so the from-scratch baselines converge cleanly. The defense fails, and in the strongest possible way (Table 3): from-scratch physics is *more* damaging than post-hoc grafting. On uniform motion, the physics penalty grows from $+0.035$ (post-hoc) to $+0.558$ (from-scratch)—the cleaner the baseline, the larger the harm. All three domains show positive penalties in both regimes; the "inject at pretraining" escape hatch is closed.

	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)
Free-rollout baseline	0.131	0.127	0.393
Fixed slot (weight 1)	0.183	0.156	0.651
Fixed slot + LBR	0.114	0.160	0.596
Probe (λ=1, K=2)	0.167	0.115^{n}	0.647
Kinematic (free MLP)	0.155	0.178	0.560
Kinematic (strict a=g)	0.206	0.173	0.559
Consistency[c] (form-free)	0.151	0.147	0.57–0.64
Label-free prior	0.171	0.172	0.653
Grounded prior	0.166	0.156	0.525
Slot+probe+LBR	0.125	0.127	0.607
Slot pos+vel, +LBR	0.207	0.093^{n}	0.621

Table 2: Post-hoc physical-structure injection, hardest clean-split latent nMSE (↓): both-OOD for uniform/collision; r/m-OOD for parabola, whose both-OOD aggregate is invalidated by the denominator pathology of §6. 28 of 30 cells fail to improve on the baseline; the two exceptions are discussed in the text. LBR = load-bearing reweighting (§4.3). [c]Consistency operates on the reweighted slot by construction. [n]Single-seed values at or below the baseline's seed range on this split only; see text. Cells are single-seed; the baseline and the LBR slot are seed-replicated in Tables 5 and 4.

Domain	Split	phys. off	phys. on	Δ
uniform	scratch	0.192	0.750	**+0.558**
	post-hoc	0.131	0.166	+0.035
parabola (r/m-OOD)	scratch	0.343	0.375	+0.03
	post-hoc	0.124	0.198	+0.07
collision	scratch	0.538	0.635	+0.097
	post-hoc	0.393	0.525	+0.132

Table 3: Pretraining vs. post-hoc physics injection (latent nMSE ↓, hardest clean split; strict a=g head + fixed slot). Physics hurts in every cell, and most where the baseline is cleanest. Parabola is judged on r/m-OOD due to the denominator pathology of its both-OOD aggregate (§6).

4.3 The Load-Bearing Problem

Why does structure fail so uniformly? The physical slot occupies 2 of 192 dimensions; under a mean prediction loss it receives $\sim$1% of the gradient, so the predictor pays almost nothing for getting it wrong. Meanwhile the black-box channel *redundantly encodes position*: even in baseline models with no slot at all, a ridge probe reads position from the latent at ρ>0.9, so a dedicated slot adds nothing the predictor does not already have, and prediction and decoding can route around it entirely. The slot aligns almost perfectly with true position (its supervision loss falls from 2.53 to 0.015) while contributing nothing: it is decodable but not load-bearing. Worse, the auxiliary gradients actively compete with prediction: at probe weight λ=10 the probe-to-prediction gradient ratio reaches 15–125$\times$ across domains and the encoder's effective dimensionality collapses by 39–90%.

Configuration (uniform)	both-OOD nMSE	seeds
Free-rollout baseline	0.136 ± 0.009	3
Fixed slot, weight 1	0.183	1
Fixed slot + LBR (w=30)	0.132 ± 0.017	3
Fixed slot + LBR (w=100)	0.136	1

Table 4: Load-bearing reweighting (LBR) restores parity, not gains: the mechanism's falsifiable prediction is confirmed—making the slot load-bearing removes its harm—but structure yields no net OOD improvement.

This account makes a falsifiable prediction: re-weighting the prediction loss so the slot dimensions carry weight (load-bearing reweighting, LBR; weight $\sim$30 on the slot) should remove the damage. On uniform motion it does (Table 4): the harmful fixed slot (0.183) returns exactly to baseline parity (0.132 ± 0.017 vs. 0.136 ± 0.009 over three seeds)—and no further. The rescue is domain-limited: on parabola and collision even the reweighted slot remains harmful (0.160 vs. 0.127; 0.596 vs. 0.393), where the slot's fixed content competes with richer dynamics. The surviving benefits are narrow and metric-dependent: with a kinematic head on the now load-bearing slot, velocity-OOD decoded position reaches ρ=0.99 and long-horizon pixel accuracy improves by $+1.25$ dB on uniform (single-seed), only in smooth domains, and at the cost of degraded zero-shot transfer (0.551, the worst of all variants). Structure, even when made load-bearing, buys interpretability rather than net robustness—consistent with attributing the reported success of extrinsic architectures (Mao, Umasudhan, and Ruchkin 2024) to their *architecture* (physical state as the sole, unavoidable prediction pathway) rather than to physical equations.

5 What Actually Helps: Training and Data

5.1 Free Rollout Dominates

Replacing teacher forcing with free rollout is the single largest and most portable improvement we find (Table 5): 2.2× on uniform, 3.6× on parabola, and 2.4× on collision, with three-seed intervals that do not overlap in any domain. The diagnosis matches the classic exposure-bias account (Bengio et al. 2015; Venkatraman, Hebert, and Bagnell 2015): under teacher forcing, one-step predictions are near-perfect (cosine 0.98–0.99) while multi-step rollouts drift at a rate ordered by dynamics complexity (uniform < parabola < collision)—the failure is invisible at training time and catastrophic at rollout time. The result carries to photorealistic simulation: on Physion++, free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for every physics-augmented variant), and among all synthetic-training configurations it transfers best to Physion OCP.

5.2 Match the Horizon to Dynamics Complexity

The rollout horizon H is not a free win; it must match the domain. Collision improves markedly at $H{\approx}20$ (both-OOD $0.393 \rightarrow 0.294$; horizon-28 cosine $0.58 \rightarrow 0.70$)—the training window must contain post-impulse futures for the model to learn them. Smooth domains show the opposite: at H=16, uniform's long-horizon cosine drops from 0.969 to 0.814 and parabola degrades on every partition; excess horizon destabilizes amplitudes where H=8 already spans the dynamics. On photorealistic video the appetite is larger still, but horizon alone is not the whole story. Longer horizons monotonically improve mid-range accuracy (Physion++ horizon-32 nMSE 0.140 at H=8 down to 0.033 at H=20); at long range, however, a plain long rollout can overshoot amplitude on some scenes (friction-collision nMSE 24.6 at cosine 0.99 under H=20 without augmentation). Pairing the long horizon with geometric scale jitter, which stabilizes amplitude, is what unlocks the regime: horizon-64 nMSE falls from 0.280 (H=8) to 0.087 (H=20+scale) and 0.014 (H=28+scale), a 19× reduction with no inflection yet found. The practical law: extend the training horizon to cover the domain's causal event span, and add geometric augmentation to keep long rollouts stable.

	teacher forcing	free rollout	gain
uniform	0.300 ± 0.007	$\mathbf{0.136 \pm 0.009}$	2.2×
parabola (r/m-OOD)	0.443 ± 0.048	$\mathbf{0.122 \pm 0.007}$	3.6×
collision	1.152 ± 0.048	$\mathbf{0.479 \pm 0.079}$	2.4×

Table 5: Headline result: free rollout vs. teacher forcing, hardest clean split, latent nMSE (↓), three seeds each (mean±sample std). Intervals do not overlap in any domain.

5.3 Augmentation: Strongest Synthetic Lever, Reversed on Photorealistic Video

On synthetic domains, simple video-only augmentation beats every physical-structure method. Per-trial appearance jitter (brightness/contrast, strength 0.5) targets the appearance shift induced by radius/mass changes and roughly halves the hardest split: uniform $0.131 \rightarrow 0.068$ (-48%), parabola -63% (with the clean r/m-OOD partition confirming, $0.127 \rightarrow 0.065$). Geometric scale jitter targets object size directly and, *only* in combination with a long horizon, sets the collision record ($0.393 \rightarrow 0.172 \pm 0.004$ over three seeds). The combinations are non-monotone and must be chosen, not stacked: appearance conflicts with long-horizon training (0.472, worse than either alone), scale synergizes with it (0.208), and the triple combination is worse than the best pair (0.253). Velocity-OOD resists all of it—temporal-stride augmentation, the natural candidate, loses even to appearance jitter (0.556 vs. 0.376)—marking unseen velocities as the open hard case.

The appearance recipe does not survive contact with photorealistic simulation. On Physion++, the same appearance augmentation that halves simple-synthetic OOD error *degrades friction-collision prediction by two orders of magnitude* (nMSE $0.062 \rightarrow 6.44$) while the *aggregate* long-horizon cosine improves—a preview of the metric traps in §6—and it does not help zero-shot Physion transfer either (0.597, below the 0.607 random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld's simple renderer but breaks in photorealistic scenes, where appearance *carries* physical information—friction, mass, and material are read off surface appearance. The split runs within augmentation itself: geometric scale jitter, which assumes only size-invariance, remains helpful at photorealistic scale (it is the long-horizon stabilizer of §5.2), whereas appearance jitter reverses. Augmentation gains are thus domain- and type-specific, not portable.

6 Four Evaluation Traps

Our anatomy repeatedly reached conclusions opposite to those suggested by the metrics most common in this literature. We document four failure modes; each reversed at least one finding in this paper.

Trap 1: cosine and probe metrics are duals of the training loss. Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from ρ=0.44 to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to 100×. Judging either result by direction-only or readout metrics inverts the conclusion.

Trap 2: nMSE denominators can degenerate. The scale-aware metric has its own failure: when the target sequence degenerates (a parabola ball exits the frame, target variance →0), per-horizon nMSE explodes—up to 2×10^6 at horizon 28 while cosine sits at a healthy 0.91—and pollutes every aggregate that includes late horizons. All six of our parabola baseline arms exhibit this blow-up, which is why parabola is judged on its clean r/m-OOD partition throughout this paper. The two traps point in opposite directions; neither metric family is safe alone.

Trap 3: zero-shot transfer is bounded by the architecture prior. On Physion OCP, a *randomly initialized* encoder scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (load-bearing weighting: 0.551), and training longer monotonically approaches—never crosses—the ceiling. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

Trap 4: protocol confounds fabricate results in both directions. Three probe-protocol choices (random vs. pretrained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive protocol stack reads uniform velocity decodability at R^2=0.166, while correcting initialization, probe window, and probe point yields 0.939—a deficit manufactured entirely by protocol. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded; 192 dropped by a parameter-renaming mismatch) invalidated an entire 45-configuration sweep before being caught by a loading guard. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pretrained model's prediction loss (0.006 vs. 0.005) while its rollout OOD error is 2.8× worse.

Checklist. (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe ρ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder.

7 Discussion and Conclusion

We asked whether physical inductive biases make latent world models obey physics, and answered with a controlled anatomy: across five injection mechanisms, two training regimes, three synthetic domains, and two photorealistic-simulation benchmarks, physical structure consistently harms OOD and long-horizon prediction—even the exact gravitational form, even with perfect labels, and *more* when co-trained from scratch. The failure has a mechanism, the load-bearing problem: information that is present in a shared latent is not thereby used by prediction, and structure grafted onto dimensions that carry no gradient weight decorates rather than constrains. The mechanism's falsifiable prediction held—reweighting removes the harm—but restores only parity. What moves physical robustness lies in training and data: free rollout (2.2–3.6×, seed-robust, on synthetic and photorealistic scenes alike), horizons matched to dynamics complexity (up to 19× on long-horizon photorealistic video when geometric augmentation stabilizes amplitude), and augmentation matched to the domain's true nuisances—with an explicit simple-synthetic-to-photorealistic reversal boundary, where appearance jitter flips from strongest lever to catastrophic while geometric jitter keeps helping. Alongside, we distilled four evaluation traps that had made structure look helpful and augmentation look safe.

The constructive reading is architectural. Our results do not falsify physically structured world models in general; they falsify *grafting* structure onto a shared latent. Extrinsic designs—where a low-dimensional physical state is the sole pathway that prediction and decoding must route through (Mao, Umasudhan, and Ruchkin 2024)—make physics load-bearing by construction, and our analysis predicts that this, not the physical equations themselves, is the source of their reported benefits. Testing that prediction by converting a shared-latent JEPA model to an extrinsic factorization is the direct next step; conservation-projection constraints that operate without labels are a complementary route for domains where states are unobservable.

Limitations. Our anatomy runs on one compact backbone (∼10M-parameter ViT-tiny JEPA); a frozen-encoder study across seven encoders up to 749M parameters supports the representational conclusions, but the training-dynamics conclusions should be re-verified at scale. Headline comparisons, the reweighting result, and the collision augmentation record are three-seed; the remaining anatomy cells are single-seed, consistently signed across domains. The synthetic domains are single-mechanism by design—that is what makes the factorial clean—and the hardest open cases we report honestly: velocity-OOD resists every augmentation we tried, and deformable-object scenes remain the weakest photorealistic-simulation category for all methods. Collision is judged in latent space only, as its pixel decoder is reconstruction-limited.

Reproducibility. All training configurations, evaluation protocols, per-run logs, and the exact numbers behind every table are documented; code and data splits build on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder calibration controls.

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A Training and Evaluation Details

Hyperparameters. All PhyWorld runs: AdamW, lr 5×10^{-5} (from-scratch reruns: 2×10^{-5}), weight decay 10^{-3}, bf16, gradient clip 1.0, 20 epochs (60 for from-scratch cells; the pretraining-regime 2×2 uses 60 epochs uniformly), batch size 64 for free rollout (H=8), 128 for teacher forcing (H=1), 48 for $H\geq16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Seeds: 3072/1234/42 where seed-replicated. Physion++ runs use the readout split (800 clips) with the same optimizer.

OOD partitions. ID: radius/mass $\in[0.7, 1.5]$, velocity $\in[1.0, 4.0]$. OOD draws radius/mass from $[0.3, 0.6]\cup[1.5, 2.0]$ and velocity from $[0, 0.8]\cup[4.5, 6.0]$. Collision uses the four-column variant (two objects' masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions ($\geq$1,600 trajectories per domain).

Probe protocol. Ridge regression (α=1), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained "paper" initialization, and K=4 stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads ρ=0.166 for uniform velocity where the corrected protocol reads 0.939. Random-encoder and pixel-statistics controls are run alongside every probe.

B Full Injection Results

Probe/slot factorial (uniform, free rollout). Latent both-OOD nMSE / pixel both-OOD PSNR (dB): baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+LBR 0.114 / 21.30; probe+slot+LBR 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

Slot reweighting sweep (uniform). Slot weight 1/30/100: 0.183 / 0.114 / 0.136 (single seed), with the w=30 cell seed-replicated at 0.132 ± 0.017 against the baseline 0.136 ± 0.009. Kinematic head on the load-bearing slot: velocity-OOD decoded position ρ=0.991/0.987 (x/y); uniform pixel horizon-28 23.34 vs. 22.09 dB (+1.25); appearance-OOD remains the weak axis (decoded ρ $0.29\rightarrow0.43$, vs. 0.96 without the kinematic head).

Augmentation interactions (collision, both-OOD nMSE). Long horizon (H=20) alone 0.294; appearance alone 0.301; appearance$\times H$=20 0.472 (conflict); scale$\times H$=20 0.208 (synergy; three seeds 0.208/0.198/0.244); appearance$\times$scale 0.262; triple 0.253; scale strength 0.5 with H=20: 0.172. Temporal-stride (velocity) augmentation: v-OOD 0.556, worse than appearance's 0.376.

Physion++ horizons. Overall nMSE at horizons 16/32/64: baseline (H=8) 0.016/0.140/0.280; H=20 0.022/0.033/0.220; H=16+scale 0.010/0.035/0.136; H=20+scale 0.012/0.024/0.087; H=28+scale reaches horizon-64 nMSE 0.014. Scale acts as an amplitude stabilizer for long rollouts: without it, H=20 explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

C Metric Pathologies

Parabola denominator blow-up. At horizons near 28, trajectories whose ball exits the frame leave near-constant target latents; the nMSE normalizer (target variance) approaches zero and per-horizon nMSE reaches 3×10^4–2×10^6 across all six parabola baseline arms while cosine stays in $[0.55, 0.95]$. Both-OOD aggregates over horizons inherit the blow-up, which is why parabola is judged on r/m-OOD (no degenerate targets) throughout.

Gradient competition. At probe weight λ=10, probe-to-prediction gradient-norm ratios are 15$\times$ (collision), 44$\times$ (uniform), and 125$\times$ (parabola); encoder effective (intrinsic) dimensionality collapses by 39–90% relative to baseline. At λ=1 the ratios are benign, which is the setting used in §4.

Transfer ceiling. Physion OCP mean AUC: random-init encoder 0.607; best synthetic-trained (free rollout) 0.603; physics-structured variants 0.551–0.582; appearance augmentation 0.579–0.597. Per-scenario, only Support (+0.10) and Collide (+0.07) show genuine signal above the random control; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing it.